1. Is there a fault in the program? If so, locate it and explain.

Yes, there is a fault. The fault is in the line with for (int i =1; i< attendance_records.size(); ++i). The loop starts with i=1 so the first lecture of the attendance record gets skipped. This can be fixed by using i=0.

2. Define a test case that does not execute the fault. Start from this question, if it is possible, you should give the test case (the attendance record with 10 values), the expected output and the actual output. If impossible, explains the reason. Besides, you should implement the complete program, including the functions under test, and verify your inputs along with their corresponding outputs.

There is no test case that doesn't execute the fault, as it is in the for loop it will always run starting at i=1. It will run for every call of the function even if an empty vector is used.

3. Define a test case that executes the fault but does not result in an error State.

Technically such a case is not possible, but if the state is only determined by the absence_count then if the first lecture is attended no error would occur because the number of absences would remain the same whether the loop started at i=1 or i=0. For example, say the array looked like {1,0,1,0,1,1,1,1,1,1} the number or absences would be the same whether lecture 1 is included or not.

4. Define a test case that results in an error state but not a failure.

Say the array was {0,0,0,0,1,1,1,1,1,1},the expected absence count is 4 but with the fault the count will be 3. However both results will result in the correct output of a class fail, so no failure in the program occurred because it's the expected output overall.

5. Define a test case that results in failure.

Say the array was {0,0,0,1,1,1,1,1,1,1},the expected absence count is 3 but with the fault the count will be 2. This will result in a failure because the output should be a fail but will instead be a pass because the absence count will not result in three or more with the fault.